

# Cube Mathematical Domain Model

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## Cube Domain Model & Coordinate Geometry

Product: Cubyntra  
Organization: Necookie Labs  
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### 1. Domain Standards & Canonical Orientation

Cubyntra adopts standard international World Cube Association (WCA) and Singmaster notation standards.

#### 1.1 Canonical Center Color Scheme (Western Standard)

In canonical orientation:

- Up (U): White (Top)
- Front (F): Green (Front)
- Right (R): Red (Right)
- Down (D): Yellow (Bottom)
- Left (L): Orange (Left)
- Back (B): Blue (Back)

Opposing pairs:

- White  $\longleftrightarrow$  Yellow (U  $\longleftrightarrow$  D)
- Green  $\longleftrightarrow$  Blue (F  $\longleftrightarrow$  B)
- Red  $\longleftrightarrow$  Orange (R  $\longleftrightarrow$  L)

### 2. 54-Facelet Layout & Indexing

Each face contains 9 stickers indexed row-major from 0 to 8:



| 6 | 7 | 8 |  
+---+---+---+

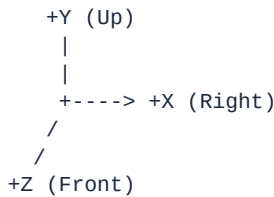
Sticker index 4 on every face is the fixed center piece:

- U[4] = 'white'
- R[4] = 'red'
- F[4] = 'green'
- D[4] = 'yellow'
- L[4] = 'orange'
- B[4] = 'blue'

### 3. 3D Cubie Coordinate System

The 3D digital twin models the cube as 27 discrete cubies positioned in a right-handed Cartesian coordinate system:

x, y, z in {-1, 0, 1}



- +X: Right face (Red)
- -X: Left face (Orange)
- +Y: Up face (White)
- -Y: Down face (Yellow)
- +Z: Front face (Green)
- -Z: Back face (Blue)

#### Cubie Classification:

- Core Cubie (1 piece): (0, 0, 0) — Internal structural core; no exterior stickers.
- Center Cubies (6 pieces): Exactly one coordinate has magnitude 1 (e.g. (0, 1, 0) -> U center).
- Edge Cubies (12 pieces): Exactly two coordinates have magnitude 1 (e.g. (0, 1, 1) -> UF edge).
- Corner Cubies (8 pieces): All three coordinates have magnitude 1 (e.g. (1, 1, 1) -> UFR corner).

### 4. Facelet-to-Cubie Spatial Mapping

| Face | Coordinate Axis | Facelet Index                                               | Cubie Coordinates (x, y, z)                                                                                          |
|------|-----------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| U    | +Y (y = 1)      | 0 \dots 2 (Row 0)<br>3 \dots 5 (Row 1)<br>6 \dots 8 (Row 2) | (-1, 1, -1), (0, 1, -1), (1, 1, -1)<br>(-1, 1, 0), (0, 1, 0), (1, 1, 0)<br>(-1, 1, 1), (0, 1, 1), (1, 1, 1)          |
| D    | -Y (y = -1)     | 0 \dots 2 (Row 0)<br>3 \dots 5 (Row 1)<br>6 \dots 8 (Row 2) | (-1, -1, 1), (0, -1, 1), (1, -1, 1)<br>(-1, -1, 0), (0, -1, 0), (1, -1, 0)<br>(-1, -1, -1), (0, -1, -1), (1, -1, -1) |
| F    | +Z (z = 1)      | 0 \dots 2 (Row 0)<br>3 \dots 5 (Row 1)<br>6 \dots 8 (Row 2) | (-1, 1, 1), (0, 1, 1), (1, 1, 1)<br>(-1, 0, 1), (0, 0, 1), (1, 0, 1)<br>(-1, -1, 1), (0, -1, 1), (1, -1, 1)          |
| B    | -Z (z = -1)     | 0 \dots 2 (Row 0)<br>3 \dots 5 (Row 1)<br>6 \dots 8 (Row 2) | (1, 1, -1), (0, 1, -1), (-1, 1, -1)<br>(1, 0, -1), (0, 0, -1), (-1, 0, -1)<br>(1, -1, -1), (0, -1, -1), (-1, -1, -1) |
| R    | +X (x = 1)      | 0 \dots 2 (Row 0)<br>3 \dots 5 (Row 1)<br>6 \dots 8 (Row 2) | (1, 1, 1), (1, 1, 0), (1, 1, -1)<br>(1, 0, 1), (1, 0, 0), (1, 0, -1)<br>(1, -1, 1), (1, -1, 0), (1, -1, -1)          |
| L    | -X (x = -1)     | 0 \dots 2 (Row 0)<br>3 \dots 5 (Row 1)<br>6 \dots 8 (Row 2) | (-1, 1, 1), (-1, 1, 0), (-1, 1, -1)<br>(-1, 0, 1), (-1, 0, 0), (-1, 0, -1)<br>(-1, -1, 1), (-1, -1, 0), (-1, -1, -1) |

### 5. Face Rotation Permutations

When a face rotates 90° Clockwise (CW), two transformations occur simultaneously:

- The 3x3 face matrix rotates 90° CW:  
$$\begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} \longrightarrow \begin{bmatrix} 6 & 3 & 0 \\ 7 & 4 & 1 \\ 8 & 5 & 2 \end{bmatrix}$$
- The 4 adjacent face strips cycle 3 stickers each:

#### Example: U Turn (Clockwise around +Y)

- $F[0, 1, 2] \longrightarrow L[0, 1, 2]$
- $L[0, 1, 2] \longrightarrow B[0, 1, 2]$
- $B[0, 1, 2] \longrightarrow R[0, 1, 2]$
- $R[0, 1, 2] \longrightarrow F[0, 1, 2]$

#### Example: R Turn (Clockwise around +X)

- $U[2, 5, 8] \longrightarrow B[6, 3, 0]$  (reversed)
- $B[6, 3, 0] \longrightarrow D[2, 5, 8]$
- $D[2, 5, 8] \longrightarrow F[2, 5, 8]$
- $F[2, 5, 8] \longrightarrow U[2, 5, 8]$

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## 6. Mathematical Invariants & Verification

The Rubik's Cube group has order:

$$|G| = \frac{8! \times 3^8 \times 12! \times 2^{12}}{2 \times 3 \times 2} = 43,252,003,274,489,856,000$$

Cubyntra validates all three invariant constraints:

- Corner Orientation Parity: Each corner has a twist in  $\{0, 1, 2\}$ . The total twist must sum to zero mod 3:  
$$\sum_{i=1}^8 \text{twist}_i \equiv 0 \pmod{3}$$
- Edge Orientation Parity: Each edge has a flip in  $\{0, 1\}$ . The total flips must sum to zero mod 2:  
$$\sum_{i=1}^{12} \text{flip}_i \equiv 0 \pmod{2}$$
- Permutation Parity: The permutation sign of the 8 corners must match the permutation sign of the 12 edges:  
$$\text{sgn}(\sigma_{\text{corners}}) = \text{sgn}(\sigma_{\text{edges}})$$

Any state violating these invariants is physically unsolvable without disassembly.